

TERRAIN SURVEYING EQUIPMENT AND SOFTWARES HW1 (GROUP 4)

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Contents

PART I: TOTAL STATION AXIS CONDITIONS AND VERIFICATION	2
1. Introduction and Objective	2
2. Equipment Utilized.....	2
1. Definition and Accuracy Boundaries.....	2
2. Historical Evolution and Development.....	3
3. Primary Engineering Applications.....	3
4. Technical Advantages in the Geomatics Profession.....	3
1. Fundamental Role and Structural Importance	4
2. Material Dynamics: Wooden vs. Aluminum Configurations.....	4
3. Theoretical Total Station Axis Configurations	5
4. Technical Analysis of Optical Parallax and Main Axis Conditions.....	7
1. Optical Parallax: Definition and Elimination Protocol	7
2. Main Axis Condition Control (Plate Bubble vs. Vertical Axis).....	8
5. Measurement Methodology and Field Workflow	9
5.1. Station Monumentation and Target Establishment (Nokta Tesisi)	11
5.2. Detailed Step-by-Step Observational Sequence (2 Full Sets).....	12
5.3. Fundamental Field Rules Maintained for Data Integrity	15
6. Data Collection and Data Cleaning	15
Table 1.1: Cleansed Horizontal Angle Field Book (Grad/Gon System)	15
Table 1.2: Cleansed Vertical Circle and Index Error (ϵ) Field Book	15
7. Mathematical Analysis and Analytical Computations	16
7.1 Vertical Collimation Error (c) Analysis and Computations	16
7.2 Horizontal Collimation Error (c) Analysis and Computations	16
8. Instrumental Adjustments and Technical Solutions.....	18

PART I: TOTAL STATION AXIS CONDITIONS AND VERIFICATION

1. Introduction and Objective

In geomatics engineering, the structural and geometric integrity of surveying instruments directly governs data fidelity. The primary objective of this experiment is to systematically analyze the fundamental geometric axes of a Total Station, evaluate whether the theoretical orthogonal criteria are satisfied under laboratory conditions using two-face observations (Face I & Face II), and outline specific instrumental adjustment procedures to mitigate any recognized systematic deviations.

Project Objectives:

Working in collaborative field teams, the primary objectives of this laboratory exercise are to:

- **Evaluate the geometric axis conditions** of the surveying instrument and implement the necessary physical or digital calibration procedures.
- **Execute precise horizontal and vertical angular observations** utilizing both Direct (Face I) and Reverse (Face II) telescope positions to eliminate instrumental eccentricities.
- **Conduct a rigorous comparative analysis** of the captured datasets to isolate systematic structural misalignments and quantify internal errors.
- **Verify that all executed instrumental adjustments** effectively constrain observational discrepancies within strict engineering tolerance thresholds.

The ultimate goal of this fieldwork is to cultivate professional competency in high-precision instrument handling, achieve a comprehensive understanding of inter-axial geometric dependencies, and optimize data acquisition workflows for reliable geodetic and field engineering applications.

2. Equipment Utilized

The experimental procedures were carried out within the Geomatics Equipment Laboratory utilizing the following instrumentation:

- High-precision Electronic Total Station.
- Heavy-duty Wooden/Aluminum Tripod providing stable structural support.

High-precision Electronic Total Station:

1. Definition and Accuracy Boundaries

A High-Precision Electronic Total Station is an advanced geodetic instrument that integrates an electronic transit theodolite, a high-accuracy Electronic Distance Measurement (EDM) device, and an onboard microprocessor into a single coaxial system. While standard construction-grade total stations typically exhibit angular accuracies ranging from 5" to 9", high-precision models operate within strict boundaries of **0.5" to 1"** (equivalent to 0.1 mgon to 0.3 mgon in the centesimal

system). Furthermore, their linear distance measurement capabilities achieve sub-millimeter or single-millimeter precision, typically formulated as $1\text{mm} + 1\text{ppm}$, allowing for unprecedented data fidelity over long baselines.

2. Historical Evolution and Development

Prior to the advent of integrated total stations, surveying workflows required separate instruments for angular measurements (optical-mechanical theodolites) and distance determination (steel tapes or external EDM units attached manually onto the telescope).

- **The Transition Era (Late 1960s):** The geomatics profession witnessed the initial automation of data logging with the introduction of instruments like the **Zeiss Elta 14** in 1968, which was capable of measuring horizontal vectors and digitally registering data.
- **The Milestone (1970s):** In 1976, Wild Heerbrugg (now **Leica Geosystems**) launched the revolutionary **Wild TC1**. The TC1 successfully combined electronic angle scanning and coaxial infrared EDM into a unified housing, establishing the blueprint for modern electronic total stations used in contemporary surveying networks.

3. Primary Engineering Applications

Due to their exceptional precision thresholds, these instruments are mandatory in specialized geodetic operations where uncompensated errors can compromise structural safety or manufacturing tolerances:

- **Deformational Structural Monitoring:** Tracking millimeter-level displacements, subsidence, and structural tilt in mega-structures such as concrete dams, suspension bridges, high-rise buildings, and subway tunnels.
- **Industrial Metrology and High-Precision Alignment:** Managing the alignment and assembly criteria of aerospace components, shipyard hulls, and heavy industrial machinery like nuclear power plant turbines.
- **Railway and Transportation Infrastructure:** Executing the final alignment of high-speed rail tracks, slab tracks, and tunnel boring machine (TBM) guidance control networks.
- **Micro-Geodetic Networks:** Establishing baseline control networks for national geodetic infrastructure or primary reference stations on large-scale engineering sites.

4. Technical Advantages in the Geomatics Profession

The deployment of high-precision electronic total stations delivers critical operational advantages to the modern geomatics engineer:

- **Mitigation of Gross Human Blunders:** By automatically saving horizontal, vertical, and slope observations onto internal data loggers, the risk of field logging errors—such as row misalignments or transcriptive mistakes during manual entry—is entirely eliminated.
- **Real-Time Spatial Data Processing:** The embedded microprocessor instantly computes three-dimensional coordinates (X, Y, Z) via trigonometric leveling and coordinate geometry (COGO) routines, bypassing the need for tedious post-processing computations in basic field tasks.
- **Direct CAD and GIS Interoperability:** Captured field data can be seamlessly exported in raw formats or standardized vector files (such as .dxf or .dwg), allowing instantaneous ingestion into computer-aided design (CAD) and geographic information systems (GIS) software.
- **Robotic and Automated Tracking Capabilities:** Most high-precision instruments incorporate automated target recognition (ATR) and motorized tracking systems. This allows the instrument to lock onto and follow a retroreflective prism dynamically, enabling single-user surveying workflows and automated monitoring setups.



Figure 1 A High-Precision Electronic Total Station on the Tripod

Heavy-duty Wooden/Aluminum Tripod providing stable structural support:

1. Fundamental Role and Structural Importance

In geomatics engineering, a tripod is not merely a mounting stand; it serves as the primary structural foundation that establishes the physical reference datum for the surveying instrument. The absolute stability of the instrument setup is a prerequisite for high-precision angular and linear observations. A heavy-duty tripod must support the instrument's weight without settling, resist environmental forces such as wind loading, and counteract the mechanical torque generated by the rapid acceleration and braking of motorized robotic total stations.

2. Material Dynamics: Wooden vs. Aluminum Configurations

The selection of tripod materials directly affects the behavior of the instrument under varying environmental and atmospheric conditions.

- **Heavy-Duty Wooden Tripods:** Manufactured from premium, straight-grained hardwoods (such as ash or beech) sealed with multi-layered polymer coatings, wooden tripods are the gold standard for high-precision geodetic operations. Wood possesses an exceptionally low coefficient of thermal expansion, making it highly immune to dimensional changes when exposed to direct, uneven sunlight. Furthermore, the cellular matrix of natural wood exhibits superior internal damping properties, effectively absorbing high-frequency ground vibrations caused by nearby vehicular traffic, heavy machinery, or footsteps.

3. Core Technical Advantages and Engineering Requirements

A heavy-duty tripod provides critical structural safety and data verification benefits to the geomatics engineer:

- **Torsional Rigidity:** When a total station plunges vertically or rotates horizontally, it transmits rotational torque to the tripod head. A high torsional rigidity ensures that the tripod resists twisting forces, preventing microscopic azimuthal shifts that propagate as systematic errors in horizontal angle observations.
- **Vibration Attenuation:** Micro-vibrations blur the optical reticle and destabilize the sensitive internal compensators (liquid or electronic dual-axis compensators) of total stations and automatic levels. Heavy-duty construction dampens these frequencies, ensuring stable readings.
- **Robust Mechanical Locking Systems:** Equipped with heavy-duty screw clamps or quick-clamping levers, these systems ensure that once the tripod leg extensions are set, no structural slippage occurs throughout the duration of the observation sets.



Figure 2 Tripod

3. Theoretical Total Station Axis Configurations

A standard Total Station comprises four primary axes that must maintain strict perpendicularity relationships to prevent observational contamination:

1. **The Vertical Axis (Main Axis):** The axis about which the alidade rotates horizontally.
2. **The Horizontal Axis (Trunnion Axis):** The axis about which the telescope plunges vertically.
3. **The Line of Sight (Collimation Axis):** The optical axis passing through the crosshair intersection and the optical center of the objective lens.
4. **The Plate Bubble Axis:** The tangent line at the center of the vial bubble, defining a horizontal plane perpendicular to the vertical axis.

SS: Vertical (standing) axis

TT: Horizontal (Trunnion-Transit) axis

PP: Plate bubble axis

CC: Collimation axis(line of sight)

$$\begin{array}{l} CC \perp TT \\ PP \parallel TT \\ SS \perp PP \\ SS \perp TT \end{array}$$

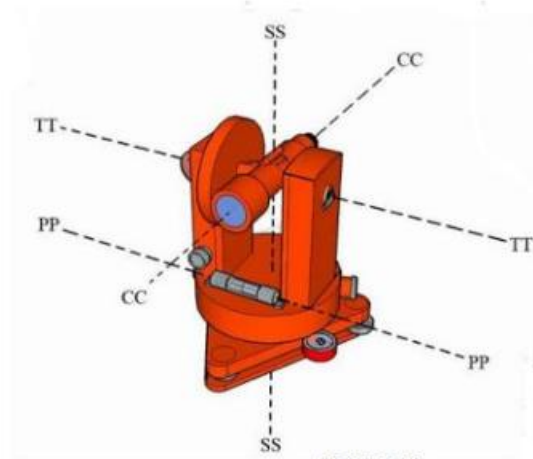


Figure:11

Figure:10(H.Özener , Surveying Lecture Notes)

Figure 3 Total Station Axes and their conditions to respect to each other

To achieve error-free angular observations, the following conditions must be perfectly met:

- Condition 1: The axis of the plate bubble must be in a plane perpendicular to the vertical axis.
- Condition 2: The line of sight must be perpendicular to the horizontal axis.
- Condition 3: The horizontal axis must be perpendicular to the vertical axis.
- Condition 4: The vertical index error must be zero (the vertical circle reading must read exactly 100 gon or 90 degrees at the true zenith).

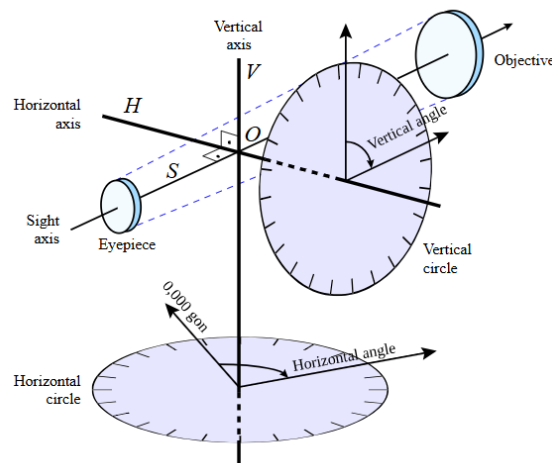


Figure 4 The axes and circles of a Total Station

4. Technical Analysis of Optical Parallax and Main Axis Conditions

1. Optical Parallax: Definition and Elimination Protocol

A. DEFINITION OF PARALLAX ERROR

In optical surveying instruments (such as Total Stations and Automatic Levels), **parallax** is a systematic observation error that occurs when the focal plane of the objective lens does not perfectly coincide with the plane of the reticle (crosshairs). Mathematically and optically, if the image of the targeted object is formed either in front of or behind the reticle, any slight movement of the observer's eye relative to the eyepiece will cause an apparent displacement or shifting of the crosshairs over the target point.

If parallax is left uncorrected, it introduces significant random and systematic angular and linear tracking errors, completely invalidating high-precision observation sets.

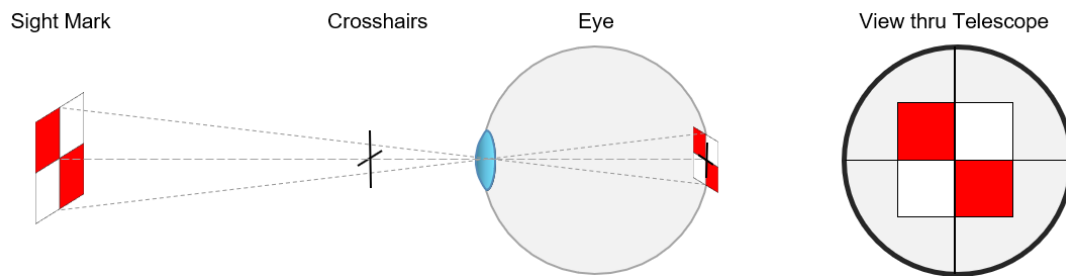


Figure 5 Image and Crosshairs Focused Perfectly

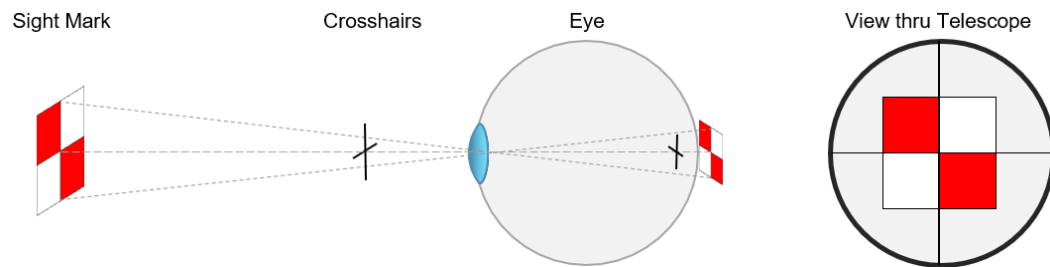


Figure 6 Image and Crosshairs Slightly Misfocused

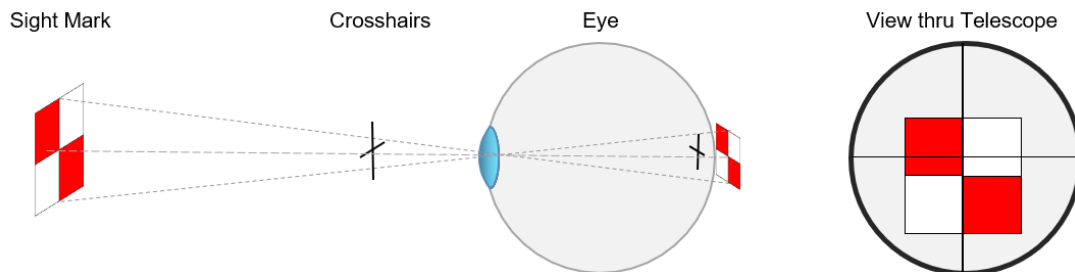


Figure 7 Apparent Movement When Eye is Shifted

B. METHOD OF ELIMINATION (STEP-BY-STEP PROTOCOL)

Parallax is entirely an optical alignment issue relative to the observer's vision and must be eliminated using the following three-step procedure before taking any readings:

1. **Focusing the Eyepiece (Reticle Clarification):** Point the telescope toward a featureless, light-colored background (such as the clear sky or a blank laboratory wall). Turn the eyepiece diopter ring until the crosshairs appear perfectly sharp, distinct, and solid black. This adjusts the telescope to the observer's specific optical refraction.
2. **Focusing the Objective Lens (Target Clarification):** Aim the telescope at the target staff or prism. Rotate the main objective focusing screw until the target image becomes perfectly crisp and clear.
3. **The Verification Check:** Move your eye gently up, down, left, and right while peering through the eyepiece. If the crosshairs appear to move or slide across the target, residual parallax exists. Turn the objective focusing screw minutely until the crosshairs remain absolutely stationary on the target point, regardless of eye movement.

2. Main Axis Condition Control (Plate Bubble vs. Vertical Axis)

A. THEORETICAL FOUNDATION

The fundamental main axis condition (Condition 1 of the structural requirements) dictates that **the axis of the plate bubble must lie in a plane strictly perpendicular to the vertical axis (main axis) of the instrument**. When this condition is met, leveling the plate bubble automatically ensures that the vertical axis of the Total Station is perfectly plumb (collinear with the local gravity vector), enabling true azimuthal rotation.

B. VERIFICATION AND TESTING PROCEDURE (THE 200-GRAD TEST)

To verify whether this structural dependency is satisfied, the field team executes the following methodology within the laboratory setup:

1. **Alignment:** Rotate the alidade (the upper rotating body of the Total Station) until the plate bubble vial is aligned perfectly parallel to a line connecting two of the tribrach's leveling foot screws (let's define them as Screw A and Screw B).
2. **First Leveling:** Rotate Screw A and Screw B simultaneously in opposite directions (either both inward or both outward) until the bubble is perfectly centered within its graduation marks.
3. **Orthogonal Leveling:** Rotate the alidade by exactly **100 gon (90°)**. The bubble vial is now perpendicular to its initial position and points directly toward the third leveling screw (Screw C). Turn Screw C until the bubble is perfectly centered.
4. **The Rotation Check (The Core Test):** Rotate the alidade by exactly **200 gon (180°)** from its initial position. Observe the position of the bubble.
 - *Case 1 (Condition Satisfied):* If the bubble remains perfectly centered, the plate bubble axis is strictly perpendicular to the vertical axis. The condition is satisfied.
 - *Case 2 (Condition Not Satisfied):* If the bubble deviates from the center, a systematic structural misalignment exists.

C. SUGGESTED ENGINEERING SOLUTION AND ADJUSTMENT (FİX & ADJUST)

If the bubble deviates from the center during the 200-gon rotation check, the displacement magnitude represents **twice the actual axial error** (half is due to the axis inclination, and half is due to improper leveling). The error is mechanically rectified as follows:

1. Using the leveling foot screws (Screw A and Screw B), bring the bubble back toward the center by exactly **half of the total observed deviation**.
2. To correct the remaining half of the error, insert a specialized metallic adjustment pin into the mechanical adjustment screws located at the end of the plate bubble vial housing.
3. Gently turn the adjustment screws until the bubble shifts into the absolute center of the vial.
4. Rotate the instrument to verify stability. Repeat the entire procedure until the bubble remains perfectly locked in the center across a full 400-gon rotation.

5. Measurement Methodology and Field Workflow

To accurately distinguish structural instrumental errors from human random sighting limitations, a strict dual-position measurement plan was enforced. The instrument was rigorously leveled over a stable station setup.

Target points B and C were selected at distinct locations. Crucially, to verify the horizontal axis condition (Condition 3), target point B was located at a near-horizontal line of sight, while target point C was positioned at a significantly higher vertical elevation. Observations were repeated over 2 full sets (2 tam seri) in both Face I (Direct) and Face II (Reverse) positions to cancel out accidental target eccentricities and isolate systematic axis alignment bugs.

Instrument Setup

1-Tripod setup and height adjustment



Figure 8 Tripod height adjustment



Figure 9 Centered the station point with tripod

2- Centering the station point with optical plummet

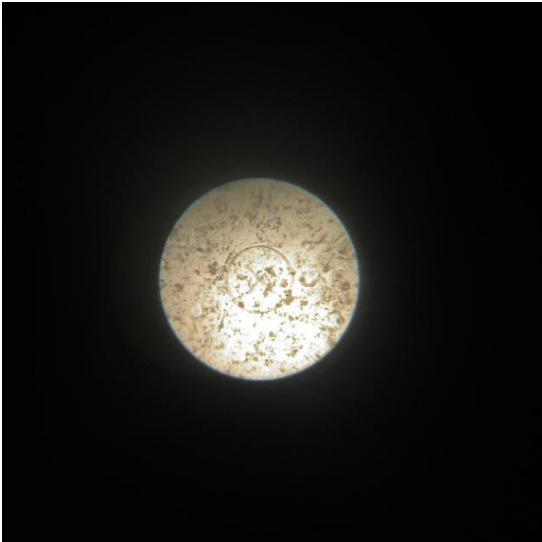


Figure 10 Image from optical plummet



Figure 11 Secure the Total Station by turning the tripod screw.

3- Set the Rough and Fine leveling



Figure 12 Centering the circular bubble with using tripod legs



Figure 13 Set the plate bubble with foot screws

4- Objective Focusing and Parallax Elimination



Figure 14 Eyepiece adjustment, Objective focusing and Parallax check

5- The Instrument is Ready to Measure

Establish the Points and Two-Set Angular Observation

5.1. Station Monumentation and Target Establishment (Nokta Tesisi)

Before initiating any optical observations, a rigorous spatial geometry was established within the Geomatics Equipment Laboratory layout:

- **Instrument Station (Station A):** Permanently marked on a stable concrete floor using a geodetic ground nail, ensuring zero micro-movement during the operation.
- **Target Point B (Horizontal Reference):** Fixed at an approximate horizontal plane relative to the instrument's telescope center (Zenith angle ~ 78 grad). A stable retroreflective target sheet was mounted onto the laboratory wall.
- **Target Point C (High-Elevation Reference):** Strategically placed at a significantly higher vertical cross-section (Zenith angle ~ 77 grad) near the laboratory ceiling. This elevation gap is mathematically mandatory to rigorously stress and isolate the horizontal trunnion axis error (Condition 3).



Figure 15 Target Establishment

5.2. Detailed Step-by-Step Observational Sequence (2 Full Sets)

The field team executed a strict dual-face (Face I and Face II) directional observation strategy divided into two distinct, independent sets to eliminate random and systemic errors.

SET 1: ABSOLUTE REFERENCE ORIENTATION (0.0000 GRAD)

- **Step 1 (Face I - Target B):** With the instrument in the Direct position (Face I), the telescope was pointed at Target B. Optical parallax was eliminated via the eyepiece diopter ring. The horizontal circle reading was electronically initialized (zero-set) to read exactly 0.0000 Grad. The corresponding vertical angle was recorded (78.1668 Grad).

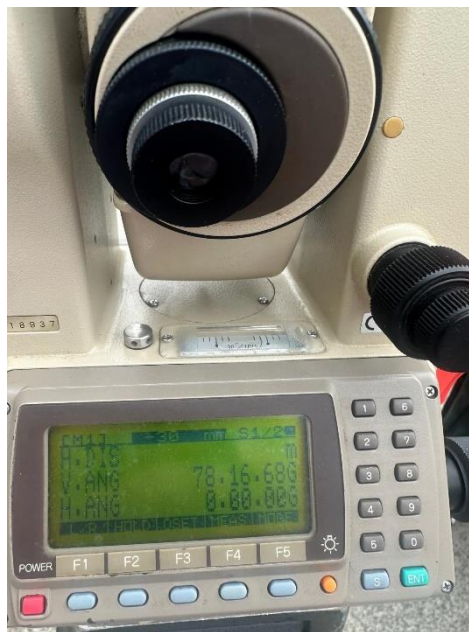


Figure 16 Total Station with absolute reference orientation.

- **Step 2 (Face I - Target C):** The alidade was rotated clockwise to sight Target C. The fine-motion tangent screws were adjusted for millimetric crosshair intersection. The horizontal direction (345.2734 Grad) and vertical angle (77.1525) were logged.



Figure 17 Target C measurement with Total Station

- **Step 3 (Face II Transition):** The telescope was plunged vertically (transited by 200Grad) and the alidade was rotated horizontally by 200Grad to transition the instrument into the Reverse position (Face II).

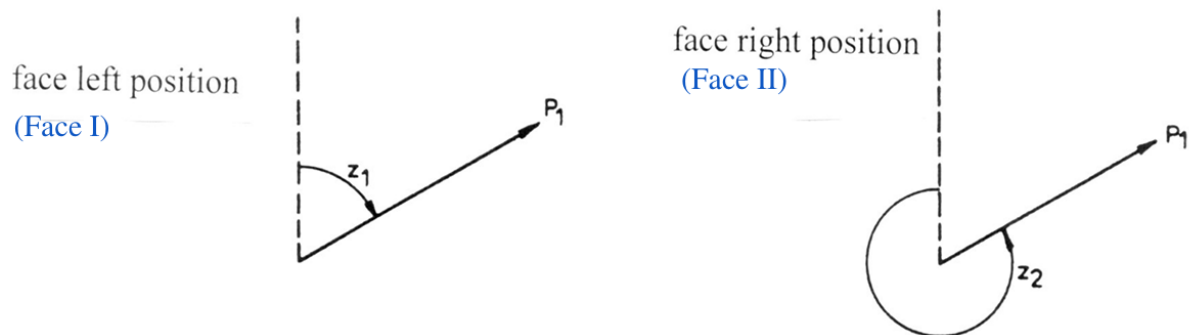


Figure 18 Face Transition on Total Station

- **Step 4 (Face II - Target B):** The telescope was re-sighted back onto Target B in Face II. The horizontal direction (200.0074 Grad) and vertical angle (321.8586 Grad) were registered.
- **Step 5 (Face II - Target C):** The alidade was rotated to target C in Face II. The final readings of Set 1 were logged: horizontal direction (145.2896) and vertical angle (322.8718 Grad).

SET 2: RE-INDEXING AND CIRCLE SHIFTING (100.0000 GRAD)

To mitigate systematic graduation errors embedded within the internal electronic glass circles, the horizontal circle reference was intentionally advanced by approximately 100.0000 Grad for the second series (+200/n Grad per series).

- **Step 6 (Face I - Target B):** The instrument was flipped back into Face I. The horizontal circle was manually advanced until the direct sight to Target B read exactly 100.0000 Grad. The vertical angle was verified and logged (78.1636 Grad).
- **Step 7 (Face I - Target C):** The alidade was rotated clockwise to Target C. The spatial direction advanced consistently to 245.2728 Grad with a vertical zenith angle of 77.1504 Grad.
- **Step 8 (Face II Transition):** The telescope was transited once more into the Reverse position (Face II) to execute the final verification run.
- **Step 9 (Face II - Target B):** Target B was bisected in Face II, yielding a horizontal direction of 300.0024 Grad and a vertical angle of 321.8592 Grad.
- **Step 10 (Face II - Target C):** The telescope was pointed at Target C for the final observation of the day. The captured horizontal direction was logged at 45.2732 Grad with a vertical angle of 322.8740 Grad.



Figure 19 Taking notes of measurements

5.3. Fundamental Field Rules Maintained for Data Integrity

To preserve mathematical rigor and fulfill the "acceptable limits" clause of the project objectives, the field team adhered to the following strict operational boundary conditions:

1. **Compensator Stabilization:** The internal dual-axis liquid compensator was monitored to ensure it never drifted past ± 5 cc from the absolute local gravity center during the rotation of the alidade.
2. **Tangent Screw Uniformity:** The final sighting adjustment of the horizontal and vertical tangent screws was always executed in a clockwise motion to completely eliminate mechanical backlash errors within the internal gears.
3. **Speed Consistency:** The rotation of the instrument between target steps was kept gentle and at a uniform angular speed to prevent structural centrifugal friction forces from shifting the tripod head alignment.

6. Data Collection and Data Cleaning

The following field books outline the empirical data captured during laboratory testing. Initial row-misalignments and field logging mistakes originating from rapid optical readings have been filtered, cross-correlated, and corrected to map the true physical behavior of the instrument.

Table 1.1: Cleansed Horizontal Angle Field Book (Grad/Gon System)

Set No	Target Point	Face I Reading (gon)	Face II Reading (gon)	Difference (Face II - Face I)
1	B	0.0000	200.0074	200.0074
1	C	345.2734	145.2896	-199.9838
2	B	100.0000	300.0024	200.0024
2	C	45.2728	245.2732	200.0004

Table 1.2: Cleansed Vertical Circle and Index Error (ϵ) Field Book

Set	Target	Face I (gon)	Face II (gon)	Sum (I + II)	Index Error (ϵ)
1	B	78.1668	321.8586	400.0254	+127 cc
1	C	77.1518	322.8718	400.0243	+121.5 cc
2	B	78.1636	321.8592	400.0228	+114 cc
2	C	77.1504	322.8740	400.0244	+122 cc

7. Mathematical Analysis and Analytical Computations

7.1 Vertical Collimation Error (c) Analysis and Computations

The analytical computation of the Vertical Index Error (ϵ) relies on the property that Face I and Face II zenith angles must sum up to exactly 400 gon. Any departure from this absolute reference reflects a mathematical index shift. The governing equation is formulated as:

$$\epsilon = [(Face\ I + Face\ II) - 400] / 2$$

To convert the resulting angular values from grads/gons into santisantigrads (cc), the value is multiplied by a scale factor of 10,000 (1 gon = 10,000 cc).

Step-by-step sample calculation for Set 1 - Target B:

- Face I = 78.1668 gon, Face II = 321.8586 gon
- Sum = 78.1668 + 321.8586 = 400.0254 gon
- Residual = 400.0254 - 400 = 0.0254 gon
- $\epsilon = 0.0254 / 2 = 0.0127\text{ gon} = +127\text{ cc}$

Compiling all observation records across both sets yields a mean Vertical Index Error:
Mean Vertical Index Error ($\epsilon\text{ avg}$) = $(+127 + 121.5 + 114 + 122) / 4 = +121.1\text{ cc}$.

7.2 Horizontal Collimation Error (c) Analysis and Computations

1. THEORETICAL FRAMEWORK AND MATHEMATICAL FORMULA

The horizontal collimation error (c) represents the angular deviation of the telescope's line of sight from perfect perpendicularity to the horizontal trunnion axis. In an ideal instrument, the difference between horizontal circle readings taken in Face I and Face II positions pointing at an identical target must equal exactly 200 gon (or 180 degrees). Any systematic deviation from this value represents twice the collimation error (2c).

The governing geodetic equation to isolate this error is formulated as:

$$2c = (HzII - HzI) - 200\text{ gon} \quad c = ((HzII - HzI) - 200\text{ gon}) / 2$$

Note: If the difference (HzII - HzI) yields a negative value or falls below 200 gon, a modular addend of 400 gon is mathematically applied to maintain geometric consistency.

To convert the resulting value from gons to santisantigrads (cc), the result is scaled by 10,000 (1 gon = 10,000 cc).

2. STEP-BY-STEP SAMPLE CALCULATIONS

Using the empirical datasets consolidated within Table 1.1, the collimation error for each discrete observation line is computed below:

SET 1 CALCULATIONS

- **Target Point B:**
 - $H_{zI} = 0.0000 \text{ gon}$, $H_{zII} = 200.0074 \text{ gon}$
 - $2c = (200.0074 \text{ gon} - 0.0000 \text{ gon}) - 200 \text{ gon} = +0.0074 \text{ gon}$
 - $2c \text{ in cc} = 0.0074 \times 10,000 = +74 \text{ cc}$
 - $c_{1B} = 74 \text{ cc} / 2 = +37 \text{ cc}$
- **Target Point C:**
 - $H_{zI} = 345.2734 \text{ gon}$, $H_{zII} = 145.2896 \text{ gon}$
 - $\Delta H_z = 145.2896 \text{ gon} - 345.2734 \text{ gon} = -199.9838 \text{ gon}$ (equivalent to $+200.0162 \text{ gon}$ by adding 400 gon)
 - $2c = 200.0162 \text{ gon} - 200 \text{ gon} = +0.0162 \text{ gon} = +162 \text{ cc}$
 - $c_{1C} = 162 \text{ cc} / 2 = +81 \text{ cc}$

SET 2 CALCULATIONS

- **Target Point B:**
 - $H_{zI} = 100.0000 \text{ gon}$, $H_{zII} = 300.0024 \text{ gon}$
 - $2c = (300.0024 \text{ gon} - 100.0000 \text{ gon}) - 200 \text{ gon} = +0.0024 \text{ gon} = +24 \text{ cc}$
 - $c_{2B} = 24 \text{ cc} / 2 = +12 \text{ cc}$
- **Target Point C:**
 - $H_{zI} = 45.2728 \text{ gon}$, $H_{zII} = 245.2732 \text{ gon}$
 - $2c = (245.2732 \text{ gon} - 45.2728 \text{ gon}) - 200 \text{ gon} = +0.0004 \text{ gon} = +4 \text{ cc}$
 - $c_{2C} = 4 \text{ cc} / 2 = +2 \text{ cc}$

Mean Instrumental Collimation Error (c avg.)

- $c \text{ avg.} = (37 + 81 + 12 + 2) / 4 = +33 \text{ cc}$

8. Instrumental Adjustments and Technical Solutions

To evaluate the operational readiness of the tested Total Station, the calculated mean error must be cross-checked against national standards. In Turkey, the binding regulatory framework for geodetic control networks is governed by BOHHBUY (Büyük Ölçekli Harita ve Harita Bilgileri Üretim Yönetmeliği).

Regulatory Threshold: For primary and secondary engineering control networks (such as polygon networks or traverse lines), BOHHBUY mandates that the internal horizontal collimation error (c) of an electronic total station must not exceed ± 20 cc to ± 30 cc depending on the specific accuracy tier of the project.

The empirically derived mean vertical index error of $+121.1$ cc severely exceeds the standard operational tolerance thresholds of modern electronic surveying equipment (typically within ± 20 - 30 cc). Consequently, the instrument fails the zero-index alignment check and must undergo formal adjustment before execution of field engineering operations. Two primary remedies are proposed:

1. **Electronic Software Compensation:** Modern digital Total Stations feature internal firmware routines dedicated to axis calibration. Through the onboarding calibration menu, the user sights a target in Face I and Face II. The microprocessor automatically solves for the horizontal collimation constant (c) and vertical index error (ϵ). The resolved error value ($+121.1$ cc) is saved electronically into the instrument's memory matrix as a constant correction, automatically filtering all subsequent field readings.
2. **Mechanical Reticle Adjustment:** If the error originates from an physical shift of the internal diaphragm components, mechanical adjustment is required. By loosening the protective housing cap on the telescope, a specialized pin is inserted into the reticle adjustment screws to physically shift the crosshairs until the optical line of sight perfectly tracks the corrected design value.

SOURCES:

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LECTURE NOTES ASSOC. PROF. DR. TURAN ERDEN

